BACH DUY DAO

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Education

Hanoi-Amsterdam High School for the Gifted — Hanoi
Grade 10 GPA: 9.2 — May 2025
Grade 11 GPA: 9.3 — May 2026

Standardized Tests
IELTS: 7.5 — Aug 2026
A Level Mathematics: A* — Aug 2026
A Level Physics: A
A Level Computer Science (self-study): A
IGCSE Mathematics: A* — Aug 2024

Honors & Awards

Gold Medal | 1I1W International Competition – Project: "Air Map – A Real-time Air Quality Monitoring and Visualization System for Hanoi." — 2026

Merit Award | FCT Summer Camp Hackathon — 2026
National Championship | VEX Robotics National Championship — 2025
Innovate Award | VEX Robotics National Championship — 2025
Official Letter of Commendation | STEM Education Promotion Alliance – Recognized for exceptional technical performance at the VEX Robotics National Championship — 2025
Bronze Medal | Vietnam AI Championship — 2025
Merit Award | Vietnam STEM, AI & Robotics Competition — 2024
Certificate of Commendation | Association for Promoting STEM Education — 2024
Gold Medal | ITMC International Talent Mathematics Competition (Round 1 in Vietnam (Nationals)) — 2024
Merit Award | VioEdu Competitive Mathematics Arena — 2023 - 2024
Computer Science Course Certification | FPT Software Academy — 2024 - 2025
Bronze Medal | 2024 Magic Music Festival — 2024
Gold Medal | AI for Good National Competition — 2026
Third Prize | USTH Future Young Scientist Competition — 2026
Gold Medal | Hong Kong International Computational Olympiad (HKICO) — 2026
Computer Science Course Certificate | MIT OpenCourseWare / edX Python Programming — 2024

Research Experience

Co-Author | Reducing Traffic Congestion in Hanoi, Vietnam Using Artificial Intelligence and Internet of Things - [Detail] - *Published in the IJMSM, Vol. 3, Issue 3, 2026* — Jul - Aug 2026

Contributions:

- Directed the end-to-end R&D of an enterprise-grade AI and IoT traffic control platform, deploying computer vision edge nodes and deep reinforcement learning models to automate city-wide urban logistics.
- Architected scalable predictive data pipelines and multi-agent network coordination protocols, mitigating bottleneck propagation and optimizing traffic throughput across complex metropolitan grids.

Leadership Experience

Head of Logistics | "Fisico 25: Floravale" Project — 2025

- Spearheaded the logistics operations, financial procurement, and venue setup schedules for the flagship event.
- Managed inventory control, resource allocation, and directed a team of volunteers to execute physical infrastructure setup under tight deadlines.
- Hosted and contributed to 20+ experiments for 100+ children.

Personal Project

Project Lead & Developer | Smart Pillow - Non-Invasive Sleep & Respiratory Monitoring System 2026

Highlights:

- Architected and built an IoT-based smart pillow prototype powered by an ESP32 microcontroller, integrating a non-invasive multi-sensor array (3× temperature, 1× humidity, and sound channels) for continuous sleep tracking without wearables.
- Implemented a fuzzy logic inference model that uses sensor gradients to accurately track core body temperature trends, head positioning, and night-sweating patterns.
- Designed end-to-end data pipelines streaming real-time metrics via BLE to a cloud dashboard, enabling automated health alerts and trend analytics for elderly care and sleep disorder monitoring.

Social Work & Activities

Committee Member (Electronics & Human Resources) | GreenAms Robotics Team (GART) & GART Expo 2025: Synthetica 2024 - 2025

- Engineered customized circuit layouts, microcontroller wiring schematics, and sensor integration for competitive VEX robots, while collaborating with the programming team to debug hardware connectivity and optimize power distribution during matches.
- Managed internal member databases and performance tracking logs, co-organized interview workflows and evaluation metrics during the annual recruitment drive, and facilitated team-building initiatives to drive team collaboration.
- Designed and installed interactive electronic displays, circuitry exhibits, and lighting networks for the main showcase arena. Also participated in a small internal robotics competition.

Logistics & Operations Committee Member | PYHA - Physics & Youth Hanoi Amsterdam Club, FISICO'25: FLORAVALE & Fairsics 2025: Viễn 2024 - 2025

- Coordinated venue preparation, materials procurement, equipment transportation, operational scheduling, and on-site event execution.
- Supported large-scale science exhibitions, seminars, and educational workshops by ensuring smooth logistics and participant experience.

Active Member | AI Young Talent Club (AIYT) 2024 - 2025

- Engaged in weekly professional research forums, artificial intelligence foundational study groups, and peer-to-peer programming workshops.

Community Service Volunteer | Bach Mai Hospital Charity Program & Warm Winter 2025 Humanitarian Campaign 2025 & 2026

- Organized and distributed meals, winter supplies, clothing, educational materials, and care packages to disadvantaged families and remote communities.
- Collaborated with volunteer teams and local organizations to support humanitarian relief efforts, contributing over 100+ service hours.

STEM & Education Volunteer | GART Camp 2025, Knowledge and Love Journey, Ngoc Vung Island Educational Outreach Program, Summer English Club 2025 (CNCTech) 2025 - 2026

- Designed and delivered robotics, STEM, programming, and English learning activities for students across multiple outreach initiatives.
- Taught robotics and computational thinking to over 50 children, facilitated interactive English workshops, and supported the development of communication and problem-solving skills.
- Contributed to educational equity by bringing STEM and language-learning opportunities to underserved communities and rural schools.